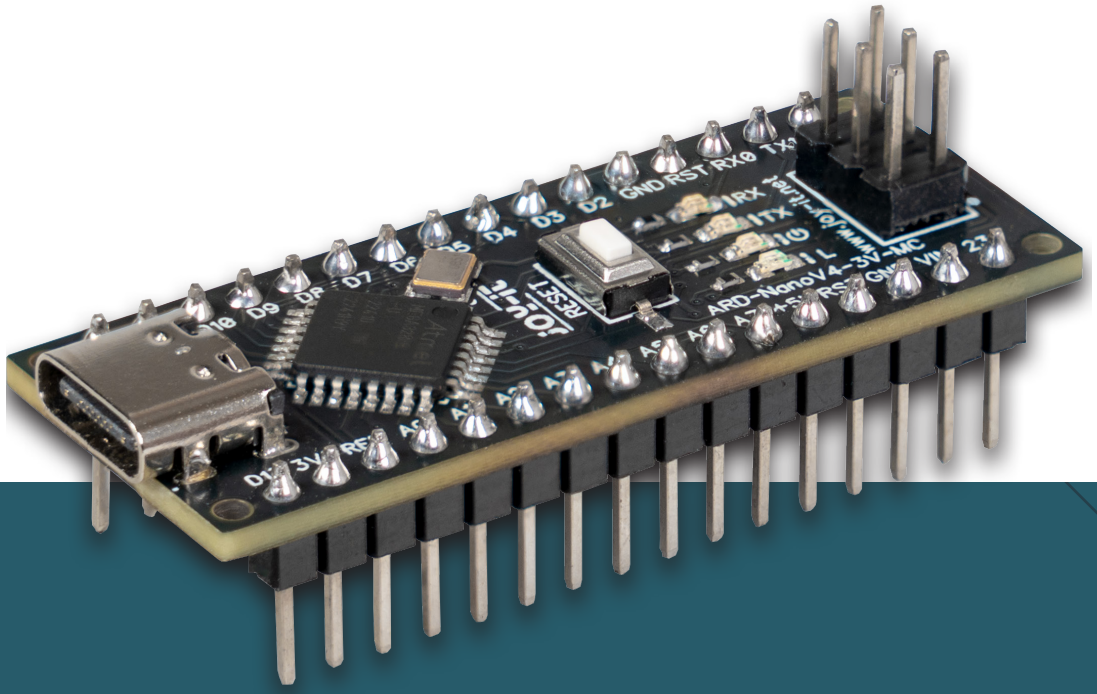


NANO V4 3V MINICORE

ARD-NanoV4-3V-MC

1. GENERAL INFORMATION

Dear customer,
thank you for choosing our product. In the following we will show you what you need to bear in mind when commissioning and using.

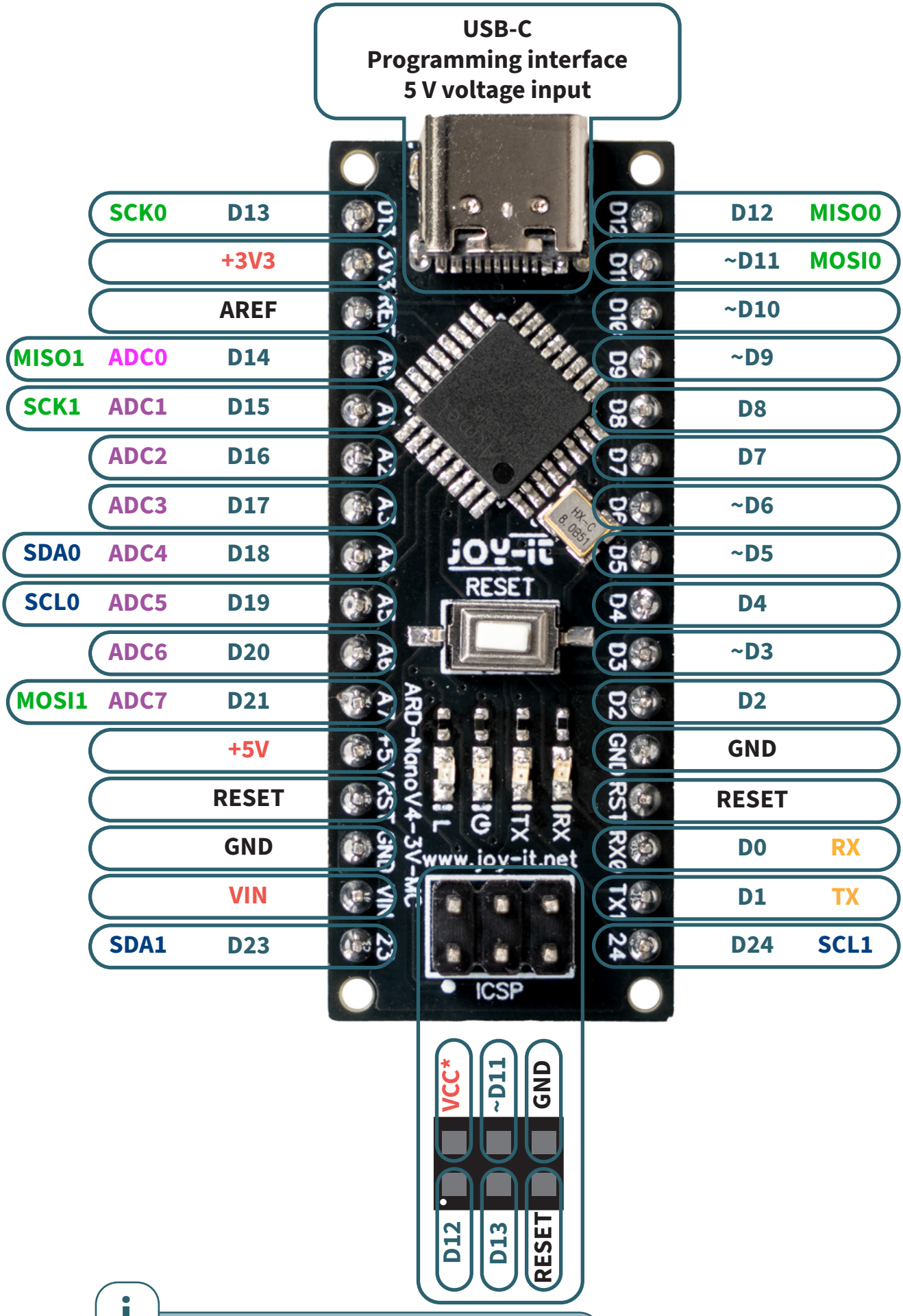
Should you encounter any unexpected problems during use, please do not hesitate to contact us.

This manual has been automatically translated.

The ARD-Nanov4-3V-MC is a particularly small microcontroller that has been specially developed for working with plug-in boards, thanks to the pin header that leads out at the bottom.
the integrated USB Type-C interface can be used to supply the circuit and board with power and to transfer programmes to the microcontroller.
The logic level, 3.3V as standard, can easily be switched to 5V via a solder bridge. The board also has a high-precision clock generator that is optimised for demanding applications and precise time measurements.

Please ensure that you use the appropriate instructions for your specific board - either ARD-NANOV4-3V or ARD-NANOV4-3V-MC. Both boards are very similar, but require different configurations of the development environment.
Using the wrong instructions will result in the board not working properly.

2. DEVICE OVERVIEW



~ PWM pins
* 3.3V/ 5V depending on solder bridge

NANO PIN	IC PORT	MAIN FUNCTION	ALTERNATIVE FUNCTION
REF	AREF	Reference voltage ADC	-
RST	PC6	Reset	PCINT14
RX0	PD0	Digital I/O / UART0 RX	RXD0, OC3A, PCINT16
TX0	PD1	Digital I/O / UART0 TX	TXD0, OC4A, PCINT17
D2	PD2	Digital I/O / Interrupt	INT0, OC3B, OC4B, PCINT18
D3	PD3	Digital I/O / PWM / Interrupt	INT1, OC2B, PCINT19
D4	PD4	Digital I/O	XCK0, T0, PCINT20
D5	PD5	Digital I/O / PWM	T1, OC0B, PCINT21
D6	PD6	Digital I/O / PWM	AIN0, OC0A, PCINT22
D7	PD7	Digital I/O	AIN1, PCINT23
D8	PB0	Digital I/O	ICP1, CLKO, PCINT0
D9	PB1	Digital I/O / PWM	OC1A, PCINT1
D10	PB2	Digital I/O / PWM / SPI0 SS	SS0, OC1B, PCINT2
D11	PB3	Digital I/O / PWM / SPI0 MOSI	MOSI0, TXD1, OC2A, PCINT3
D12	PB4	Digital I/O / SPI0 MISO	MISO0, RXD1, PCINT4
D13	PB5	Digital I/O / SPI0 SCK	SCK0, XCK0, PCINT5
A0	PC0	Digital I/O / ADC0	ADC0, MISO1, PCINT8
A1	PC1	Digital I/O / ADC1	ADC1, SCK1, PCINT9
A2	PC2	Digital I/O / ADC2	ADC2, PCINT10
A3	PC3	Digital I/O / ADC3	ADC3, PCINT11
A4	PC4	Digital I/O / ADC4 / I^2C0 SDA	ADC4, SDA0, PCINT12
A5	PC5	Digital I/O / ADC5 / I^2C0 SCL	ADC5, SCL0, PCINT13
A6	PE2	Digital I/O / ADC6 / SPI1 SS	ADC6, ICP3, SS1, PCINT26
A7	PE3	Digital I/O / ADC7 / SPI1 MOSI	ADC7, MOSI1, T3, PCINT27
D23	PE0	Digital I/O / I^2C1 SDA	ACO, ICP4, SDA1, PCINT24
D24	PE1	Digital I/O / I^2C1 SCL	TC4, SCL1, PCINT25

Further technical details regarding the individual pins, special functions, electrical limits and register mappings can be found in the official ATmega328PB datasheet.

3. LOGIC LEVEL

You can change the logic level of the board from 3.3 V to 5 V by adjusting the solder bridge on the underside of the board accordingly. This solder bridge determines the supply voltage with which the microcontroller is operated - and therefore also its logic level.

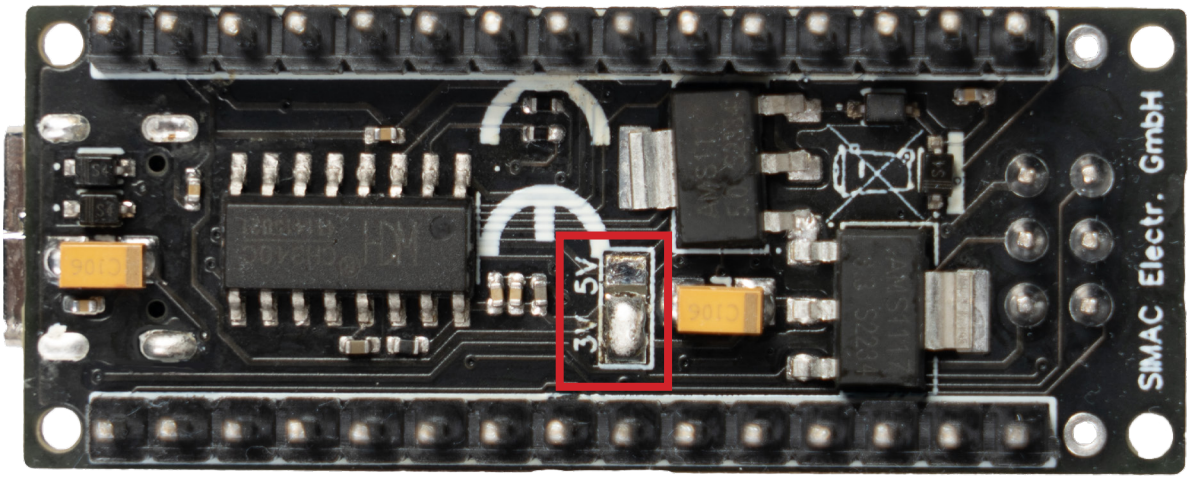
The board is set to 3.3 V at the factory. This means that the microcontroller works with a supply voltage of 3.3 V, and all input and output pins also use 3.3 V as the logic level - ideal for many modern sensors and wireless modules.

However, if you want to work with 5 V logic, you can change the voltage as follows:

- Turn the board over and look for the three-pole solder bridge labelled "3V 5V".
- Remove the existing solder that connects the centre pad to the left pad ("3V").
- Now connect the centre pad to the right-hand pad ("5V") using a new solder bridge.

This supplies the microcontroller directly with 5 V and all digital I/O pins then also work with 5 V logic levels.

Attention!!! After switching to 5 V, you should not use any components that are designed exclusively for 3.3 V, as their inputs or outputs may be damaged by the higher levels. It is therefore essential to ensure that all connected modules and sensors are 5 V-tolerant.



4. SOFTWARE SETUP

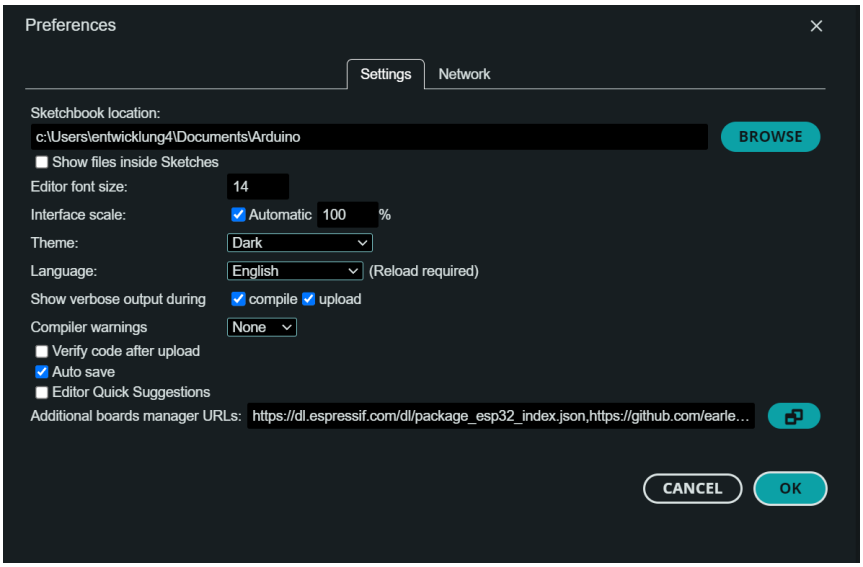
The Arduino IDE is usually used to program the board.
You can download them here:
https://www.arduino.cc/en/software

Once you have downloaded and installed the software, you can start it.

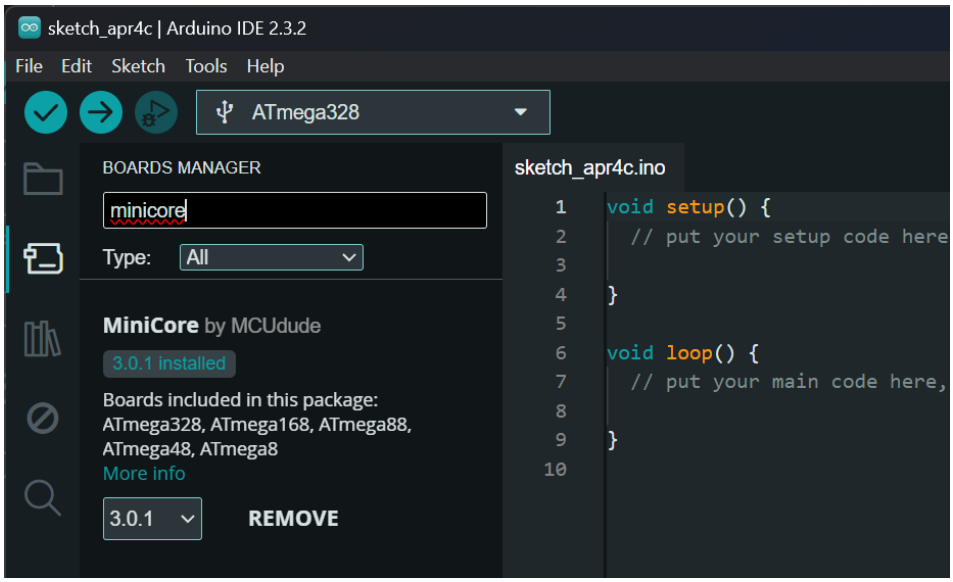
Before you can load a sketch, you need to make a few settings for the board.

First add this additional board manager URL under **File** → **Preferences**:

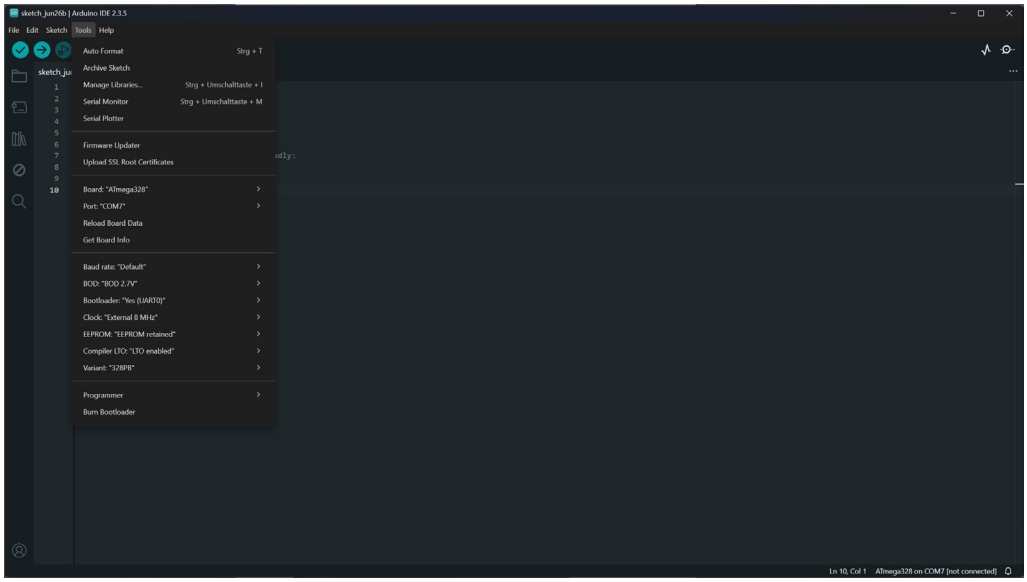
https://mcudude.github.io/MiniCore/package_MCUdude_MiniCore_index.json



Now you can search for minicore under **Tools** → **Board** → **Boards Manager**...
and install the **MiniCore** board manager from **MCUDude**.

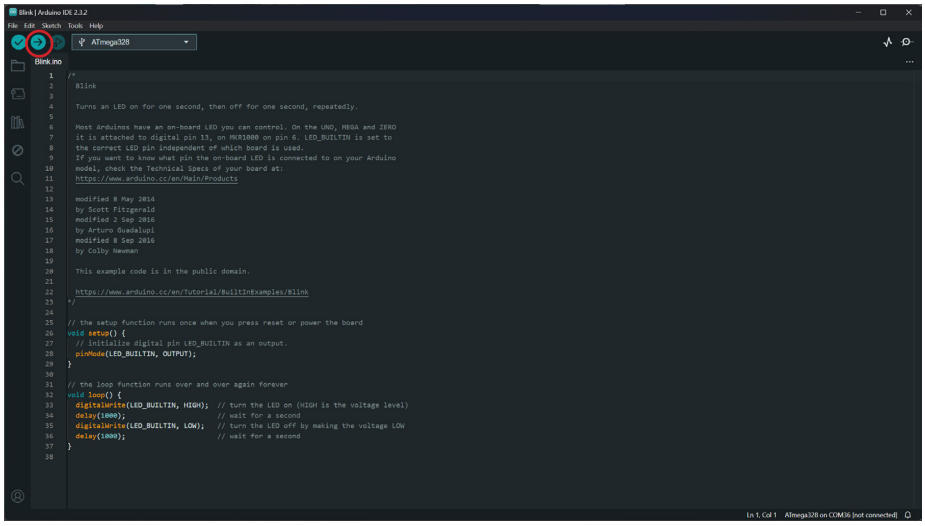


Now select the appropriate board: **Tools** → **Board** → **Minicore** → **ATmega328**
At **Tools** → **Port**, select the port to which your device is connected.
At **Tools** → **Clock**, select **External 8 MHz**.
At **Tools** → **Variant**, select **328PB**.
And at **Tools** → **Programmer** select **AVRISP mkII**



5. CODE EXAMPLE

To test your configuration, you can run a simple code example on your NanoV4.
to do this, open the file under **File** → **Examples** → **01.Basics** → **Blink**
Now upload the example by clicking on **Upload**.



This example code makes the LED on the board flash.

6. INFORMATION & TAKE-BACK OBLIGATIONS

Our information and take-back obligations under the German Electrical and Electronic Equipment Act (ElektroG)



Symbol on electrical and electronic equipment:
This crossed-out wheelie bin means that electrical and electronic appliances do not belong in household waste. You must hand in the old appliances at a collection point. Before handing in old batteries and accumulators that are not enclosed by the old appliance must be separated from it.

Return options:
As an end user, you can return your old appliance (which essentially fulfils the same function as the new appliance purchased from us) for disposal free of charge when purchasing a new appliance. Small appliances with no external dimensions greater than 25 cm can be disposed of in normal household quantities regardless of whether you have purchased a new appliance.

Possibility of return at our company location during opening hours:
SIMAC Electronics GmbH, Pascalstr. 8, D-47506 Neukirchen-Vluyn

Return option in your neighbourhood:
We will send you a parcel stamp with which you can return the device to us free of charge. To do so, please contact us by e-mail at Service@joy-it.net or by telephone.

Packaging information:
Please pack your old appliance securely for transport. If you do not have suitable packaging material or do not wish to use your own, please contact us and we will send you suitable packaging.

7. SUPPORT

We are also there for you after your purchase. Should any questions remain unanswered or problems arise, we are also available to assist you by e-mail, telephone and ticket support system.

E-mail: service@joy-it.net
Ticket system: https://support.joy-it.net
Phone: +49 (0)2845 9360 - 50

For more information, please visit our website:
www.joy-it.net

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